P04

Quality Factor and Bandwidth in Tuned Electrical Filters

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Domain DOM_ENG

Abstract

Piezoelectric quartz resonators are represented by equivalent RLC circuits. High Q factors exceeding 10^5 produce ultra-narrow passbands and stable radio frequency reference standards.

Abstract quoted from K. S. Van Dyke et al., Proceedings of the IRE (1928).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?